



Basic Emergency Care

Chest Pain

Bewar Pirasali
Lecturer

Why Chest Pain is Important

- One of the most common emergency complaints
 - May represent life-threatening pathology
 - Requires systematic and rapid assessment
 - Early intervention reduces mortality

Definition of Chest Pain

- Unpleasant sensation in the chest area
 - May be cardiac or non-cardiac
 - Always rule out serious causes first

Initial Approach – ABCDE

- Airway – ensure patency
- Breathing – assess rate, effort, SpO₂
- Circulation – pulse, BP, perfusion
- Disability – level of consciousness
- Exposure – inspect chest completely

Immediate Stabilization Measures

- Attach cardiac monitor
- IV access
- Oxygen if $\text{SpO}_2 < 94\%$
- Continuous reassessment

History Taking – SOCRATES

- S – Site
- O – Onset
- C – Character
- R – Radiation
- A – Associated symptoms
- T – Timing
- E – Exacerbating/Relieving factors
- S – Severity

High-Risk Features in History

- Central crushing pain
- Radiation to arm or jaw
- Syncope or collapse
- Severe breathlessness
- Diaphoresis
- Cardiac risk factors

Categories of Chest Pain

- 1. Cardiac
- 2. Pulmonary
- 3. Gastrointestinal
- 4. Musculoskeletal
- 5. Psychological
 - Always think: Life-threatening vs Non-life-threatening

Cardiac Causes Overview

- Acute Coronary Syndrome
- Arrhythmias
- Cardiac Tamponade
- Aortic Dissection

Acute Coronary Syndrome (ACS)

- Unstable Angina, NSTEMI, STEMI
 - Central pressure-like pain
 - Radiation to arm/jaw
 - Nausea and sweating

ACS – Immediate Emergency Management

- ABC assessment
 - Aspirin 300 mg (if no contraindication)
 - Oxygen if hypoxic
 - Sublingual GTN if SBP > 90 mmHg
 - ECG within 10 minutes
 - Urgent referral for reperfusion

Arrhythmias

- May present with chest pain or collapse
 - Tachyarrhythmia or bradyarrhythmia
 - May progress to cardiac arrest

Arrhythmias – Emergency Management

- If unstable → immediate cardioversion
 - If no pulse → CPR + AED
 - Follow ACLS protocol

Cardiac Tamponade

- Chest pain, hypotension, raised JVP
 - Muffled heart sounds (Beck's triad)
 - Life-threatening condition

Cardiac Tamponade – Management

- Support ABC
 - IV fluids temporarily
 - Urgent referral
 - Definitive: Pericardiocentesis in hospital

Aortic Dissection

- Sudden tearing chest pain
 - Radiates to back
 - Unequal BP in arms

Aortic Dissection – Management

- High-flow oxygen
 - Control BP if possible (beta-blocker)
 - Urgent transfer to specialist center

Pulmonary Causes Overview

- Pulmonary Embolism
- Pneumothorax
- Asthma
- COPD Exacerbation

Pulmonary Embolism

- Sudden pleuritic pain
- Shortness of breath
- Tachycardia
 - Risk factors: surgery, immobilization

PE – Emergency Management

- High-flow oxygen
- IV access and monitoring
- Urgent referral for anticoagulation/thrombolysis

Pneumothorax

- Air in pleural space
- Sudden unilateral chest pain
- Reduced breath sounds

Tension Pneumothorax – Emergency

- Severe breathlessness
- Hypotension
- Tracheal deviation (late sign)
- Immediate needle decompression
- Do NOT wait for imaging

Asthma Exacerbation

- Wheezing and chest tightness
- Shortness of breath
- Prolonged expiration

Asthma – Emergency Management

- High-flow oxygen
- Salbutamol nebulizer (repeat if needed)
- Add Ipratropium if severe
- Systemic corticosteroids
- Urgent referral if poor response

COPD Exacerbation

- Chronic smoker
- Increased sputum
- Worsening dyspnea

COPD – Emergency Management

- Controlled oxygen (target SpO₂ 88–92%)
 - Salbutamol nebulizer
 - Add Ipratropium
 - Systemic steroids
 - Antibiotics if infection suspected

Gastrointestinal Causes

- GERD – burning pain after meals
- Esophageal spasm – mimics cardiac pain

GI – Management

- Antacids / PPI
 - Lifestyle advice
 - Exclude cardiac causes first

Musculoskeletal Chest Pain

- Localized, reproducible tenderness
 - Worse with movement

Musculoskeletal – Management

- Simple analgesia
 - Reassurance
 - Local heat application

Anxiety / Panic Attack

- Chest tightness
 - Hyperventilation
 - Normal examination

Anxiety – Management

- Reassurance
 - Breathing control techniques
 - Exclude organic causes first

Red Flags in Chest Pain

- Ongoing severe pain
 - Collapse or syncope
 - Severe hypotension
 - Severe breathlessness

The Big 5 – Never Miss

- MI
 - Pulmonary Embolism
 - Tension Pneumothorax
 - Cardiac Tamponade
 - Aortic Dissection

Summary Algorithm

- 1. ABCDE
 - 2. Identify life-threatening causes
 - 3. Treat immediately reversible conditions
 - 4. Stabilize
 - 5. Urgent referral if unstable

Key Take-Home Messages

- Treat what kills first
 - Use structured history (SOCRATES)
 - Differentiate cardiac vs non-cardiac
 - Early management saves lives